Grade 7 Computing

Strand 1: Introduction to Computing

Lesson 2 — Health and Safety in the Use of ICT Tools

Codes: B7.1.3.1.1 – B7.1.3.1.4 **Format:** Theory Questions and Answers

Preamble (pre-lesson)

Computers help us work, but they can also hurt the body if we ignore how we sit, look, and use the room.

This lesson answers questions about:

1. **Health measures** — how to protect your body.
2. **Safety measures** — how to avoid accidents.
3. **Applying health and safety in the lab.**
4. **Checking your workstation.**

Words to know first:

- **Health** — how well your body is.
- **Safety** — keeping yourself and others free from harm.
- **Posture** — the way you sit or stand.
- **Ergonomics** — arranging furniture so the body works comfortably.
- **Workstation** — your desk, chair, and computer as one set.
- **Surge protector** — a device that protects equipment from sudden electricity spikes.

A. Health measures

Q1. What is health? **Ans.** Health is the condition of your body and mind.

Q2. What is posture? **Ans.** Posture is the way you hold your body when you sit or stand.

Q3. How should you sit when using a computer? **Ans.** You should sit with your back straight against the chair.

Q4. Where should your feet be when using a computer? **Ans.** Your feet should be flat on the floor.

Q5. How far should the computer screen be from your eyes? **Ans.** The screen should be about an arm's length from your eyes.

Q6. Where should the top of the screen be? **Ans.** The top of the screen should be at or just below eye level.

Q7. What is the 20-20-20 rule? **Ans.** The 20-20-20 rule says that every 20 minutes you should look at something 20 feet away for 20 seconds.

Q8. Why should you follow the 20-20-20 rule? **Ans.** You should follow it to reduce eye strain.

Q9. How should your wrists be when typing? **Ans.** Your wrists should be straight when typing.

Q10. Why should you take short breaks when using a computer? **Ans.** You should take short breaks to rest your eyes, back, and mind.

Q11. What happens if the room light reflects on the screen? **Ans.** Glare appears on the screen and causes eye strain.

Q12. Why should headphone volume not be too loud? **Ans.** Loud headphone volume can damage your hearing.

Q13. Why should you wash your hands before using a shared computer? **Ans.** Washing hands reduces the spread of germs on shared equipment.

Q14. Mention three health measures when using a computer. **Ans.**

- Sit with good posture.

- Follow the 20-20-20 rule.
- Keep the screen at a safe distance.

B. Safety measures

Q15. What is safety? **Ans.** Safety is keeping yourself and others free from harm.

Q16. Why should you not pull a plug by its cord? **Ans.** Pulling a plug by its cord can damage the wire and cause electric shock.

Q17. What is a surge protector? **Ans.** A surge protector is a device that protects equipment from sudden spikes of electricity.

Q18. Why should computer vents not be blocked? **Ans.** Blocked vents stop airflow and cause the computer to overheat.

Q19. Why should cables be kept tidy? **Ans.** Tidy cables prevent people from tripping and prevent damage to the wires.

Q20. Why should you not eat or drink at the workstation? **Ans.** Food and drink can spill and damage the equipment or cause electric shock.

Q21. What should you do if you see a damaged cable? **Ans.** You should report the damaged cable and not use it.

Q22. Why should you save your work often? **Ans.** Saving often prevents loss of work if the computer fails.

Q23. Should you share earbuds with a friend? **Ans.** No, you should not share earbuds because it spreads germs.

Q24. Mention four safety measures when using ICT tools. **Ans.**

- Do not eat or drink near the computer.
- Keep cables tidy.
- Do not pull plugs by the cord.
- Report damaged equipment.

C. Health and safety in the lab

Q25. What should you do before starting a lab session? **Ans.** You should check the chair, screen, and cables before starting.

Q26. What should you do while working in the lab? **Ans.** You should sit with good posture and take short breaks.

Q27. What should you do when you finish working in the lab? **Ans.** You should save and close all files and shut down the computer properly.

Q28. Why should you push your chair in when leaving? **Ans.** Pushing the chair in keeps the lab tidy and prevents accidents.

Q29. Why should emergency exits stay clear? **Ans.** Clear emergency exits allow people to leave safely during an emergency.

Q30. Why should students be trained before using lab tools? **Ans.** Training helps students use the tools correctly and avoid accidents.

D. Workstation check

Q31. What is a workstation? **Ans.** A workstation is your desk, chair, and computer used together as one set.

Q32. What is ergonomics? **Ans.** Ergonomics is arranging furniture and equipment so the body works comfortably and safely.

Q33. State four items on a workstation safety checklist. **Ans.**

- Screen at eye level.
- Chair supports the back.

- Feet flat on the floor.
- Cables tidy and not damaged.

Q34. Why does the room need good lighting? **Ans.** Good lighting reduces eye strain and glare on the screen.

Q35. How do you know if a workstation is safe? **Ans.** A workstation is safe when it passes every item on the safety checklist.

Text diagram — correct sitting posture

```
eyes ■■■■ top of screen at eye level
| /
| / screen about an arm's length away
| /
back straight
|
chair
|
feet flat on floor
```

Text diagram — workstation safety checklist

```
+---------------------------------+
| WORKSTATION CHECK |
| [ ] Screen at eye level |
| [ ] Chair supports the back |
| [ ] Feet flat on the floor |
| [ ] Cables tidy and undamaged |
| [ ] Room well lit, no glare |
| [ ] No food or drink near PC |
| [ ] Surge protector present |
| [ ] Emergency exit clear |
+---------------------------------+
```

Exercises (8 sections × 5 questions)

Exercise 1 — Health measures

1. What is health?
2. What is the 20-20-20 rule?
3. How far should the screen be from your eyes?
4. Where should the top of the screen be?
5. **[Review]** What is posture?

Answers

1. Health is the condition of your body and mind.
2. The 20-20-20 rule says that every 20 minutes you should look at something 20 feet away for 20 seconds.
3. The screen should be about an arm's length from your eyes.
4. The top of the screen should be at or just below eye level.
5. Posture is the way you hold your body when you sit or stand.

Exercise 2 — Posture and ergonomics

1. How should you sit when using a computer?
2. Where should your feet be?
3. How should your wrists be when typing?
4. What is ergonomics?

5. **[Review]** Why should you take short breaks?

Answers

1. You should sit with your back straight against the chair.
2. Your feet should be flat on the floor.
3. Your wrists should be straight when typing.
4. Ergonomics is arranging furniture and equipment so the body works comfortably and safely.
5. You should take short breaks to rest your eyes, back, and mind.

Exercise 3 — Safety measures

1. Why should you not pull a plug by its cord?
2. What is a surge protector?
3. Why should computer vents not be blocked?
4. Why should cables be kept tidy?
5. **[Review]** Why should you not eat at the workstation?

Answers

1. Pulling a plug by its cord can damage the wire and cause electric shock.
2. A surge protector is a device that protects equipment from sudden spikes of electricity.
3. Blocked vents stop airflow and cause the computer to overheat.
4. Tidy cables prevent people from tripping and prevent damage to the wires.
5. Food and drink can spill and damage the equipment or cause electric shock.

Exercise 4 — Lab rules

1. What should you do before starting a lab session?
2. What should you do while working in the lab?
3. What should you do when you finish working in the lab?
4. What should you do if you see a damaged cable?
5. **[Review]** Should you share earbuds?

Answers

1. You should check the chair, screen, and cables before starting.
2. You should sit with good posture and take short breaks.
3. You should save and close all files and shut down the computer properly.
4. You should report the damaged cable and not use it.
5. No, you should not share earbuds because it spreads germs.

Exercise 5 — Applying health and safety

1. Mention three health measures when using a computer.
2. Mention four safety measures when using ICT tools.
3. Why should you save your work often?
4. Why should you wash your hands before using a shared computer?
5. **[Review]** What does "Restart" do?

Answers

1. Sit with good posture; follow the 20-20-20 rule; keep the screen at a safe distance.
2. Do not eat or drink near the computer; keep cables tidy; do not pull plugs by the cord; report damaged equipment.
3. Saving often prevents loss of work if the computer fails.
4. Washing hands reduces the spread of germs on shared equipment.
5. Restart turns the computer off and immediately back on again.

Exercise 6 — Workstation check

1. What is a workstation?
2. State four items on a workstation safety checklist.
3. Why does the room need good lighting?
4. What is the difference between health and safety?
5. **[Review]** What is a folder?

Answers

1. A workstation is your desk, chair, and computer used together as one set.
2. Screen at eye level; chair supports the back; feet flat on the floor; cables tidy and not damaged.
3. Good lighting reduces eye strain and glare on the screen.
4. Health is about the condition of the body while safety is about keeping people free from harm.
5. A folder is a container that holds files and other folders.

Exercise 7 — Health risks

1. What happens if the room light reflects on the screen?
2. Why should headphone volume not be too loud?
3. Why should emergency exits stay clear?
4. Why should students be trained before using lab tools?
5. **[Review]** What is the keyboard shortcut for Copy?

Answers

1. Glare appears on the screen and causes eye strain.
2. Loud headphone volume can damage your hearing.
3. Clear emergency exits allow people to leave safely during an emergency.
4. Training helps students use the tools correctly and avoid accidents.
5. Ctrl + C.

Exercise 8 — Scenario and discussion

1. A student eats lunch while typing. State two risks.
2. A friend blocks the computer fan with a jacket. What do you say?
3. The screen is too low and the room is dark. How do you fix the station?
4. **[Review]** What is the 20-20-20 rule?
5. **[Review]** What is ergonomics?

Answers

1. Spills can damage the computer and crumbs can get under the keys.
2. Tell the friend that blocking the fan can make the computer overheat.
3. Raise the screen to eye level and turn on the room lights.
4. The 20-20-20 rule says that every 20 minutes you should look at something 20 feet away for 20 seconds.
5. Ergonomics is arranging furniture and equipment so the body works comfortably and safely.

Summary table

Term	One-line meaning
Posture	the way you sit or stand
Ergonomics	arranging furniture for comfort and safety
20-20-20 rule	every 20 min look 20 ft away for 20 s
Surge protector	protects equipment from electricity spikes
Workstation	desk + chair + computer as one set

NaCCA CCP B7.1.3 — Theory Questions and Answers format. Plain simple English. British notation.